PR: 4

Calculative Foundation

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Task-1 Vector Representation

- Number of students = 15
- Number of subjects = 5
- complete score matrix shape = (15, 5)
- one student's vector shape = (5,)

Student A = Arjun (row 1) A = [85, 78, 82, 74, 90]
Student B = Diya (row 2) B = [92, 88, 85, 80, 95]

Task-2 Norms

L1 Norm

Add up to absolute values of all elements of the vector

$L_1 = \sum |x_i|$

$L_1(A) = 85 + 78 + 82 + 74 + 90 = 409$

$L_1(B) = 92 + 88 + 85 + 80 + 95 = 440$

L2 Norm

square all elements of the vector, add them up, then take the square root.

$L_2 = \sqrt{(\sum x_i^2)}$

square of A : $85^2 = 7225$

$78^2 = 6084$

$82^2 = 6724$

$74^2 = 5476$

$90^2 = 8100$

$\sum A^2 = 7225 + 6084 + 6724 + 5476 + 8100 = 33609$

$||A|| = \sqrt{33609} = 183.33$

Square of B : $92^2 = 8464$

$88^2 = 7744$

$85^2 = 7225$

$80^2 = 6400$

$95^2 = 9025$

$\Sigma B^2 = 8464 + 7744 + 7225 + 6400 + 9025 = 38858$

$\|B\| = \sqrt{38858} = 197.12$

- L_1 norm = total magnitude
- L_2 norm = vector's Eulidean magnitude

Task-3 Dot product and Angle

Dot Product

Multiply the corresponding elements of the two vectors, then add all the products.

$A \cdot B = \Sigma A_i B_i$

- $85 \times 92 = 7820$
- $78 \times 88 = 6864$
- $82 \times 85 = 6970$
- $74 \times 80 = 5920$
- $90 \times 95 = 8550$

$A \cdot B = 7820 + 6864 + 6970 + 5920 + 8550 = 36124$

Angle between two vectors.

$\cos\theta = (A \cdot B) / (\|A\| \|B\|)$

$\|A\| \times \|B\| = 183.33 \times 197.12 = 36138.32$

$\cos\theta = 36124 / 36138.32 = 0.99960$

Inverse cosine apply:

$\theta = \cos^{-1}(0.99960) = 1.61°$

The angle is very small. so both students have almost the same performance pattern.

Task-4 Cross Product and Perpendicularity

Cross product is possible only for 3D vectors, so the first 3 sub are taken.

- 1st coordinate = math
- 2nd coordinate = Physics
- 3rd coordinate = chemistry

A = (85, 78, 82) B = (92, 88, 85)

$A \times B = [A_2B_3 - A_3B_2, A_3B_1 - A_1B_3, A_1B_2 - A_2B_1]$

First: $A_2B_3 - A_3B_2$	Second: $A_3B_1 - A_1B_3$	Third: $A_1B_2 - A_2B_1$
$(78 \times 85) - (82 \times 88)$	$= (82 \times 92) - (85 \times 85)$	$= (85 \times 88) - (78 \times 92)$
$6630 - 7216$	$= 7544 - 7225$	$= 7480 - 7176$
$= -586$	$= 319$	$= 304$

Final cross product: $A \times B = [-586, 319, 304]$

Verification

$A \cdot (A \times B) = (85 \times -586) + (78 \times 319) + (82 \times 304)$
$= -49810 + 24882 + 24928$
$= 0$ ✓

$B \cdot (A \times B) = (92 \times -586) + (88 \times 319) + (85 \times 304)$
$= -53912 + 28072 + 25840$
$= 0$ ✓

The dot product is 0, so the cross product is perpendicular to both vectors.

Task-5 Vector Projection (A onto B)

$$proj_B A = \left(\frac{(A \cdot B)}{\|B\|^2}\right) \times B$$

$A \cdot B = 36124$ $\quad \|B\|^2 = 38858$

$$\text{Scalar} = \frac{36124}{38858} = 0.92964$$

Projection vector.

- $X_1 = 0.92964 \times 92 = 85.53$
- $X_2 = 0.92964 \times 88 = 81.81$
- $X_3 = 0.92964 \times 85 = 79.02$
- $X_4 = 0.92964 \times 80 = 74.34$
- $X_5 = 0.92964 \times 95 = 88.32$

Proj_BA = [85.53, 81.81, 79.02, 74.34, 88.32]

Task-6 Matrix operations

A = 85, 78
B = 92, 88

$$\text{Matrix } A = \begin{bmatrix} 85 & 78 \\ 92 & 88 \end{bmatrix}$$

Matrix Addition $A + A = 2A$

$$\begin{bmatrix} 85+85 & 78+78 \\ 92+92 & 88+88 \end{bmatrix} = \begin{bmatrix} 170 & 156 \\ 184 & 176 \end{bmatrix}$$

Matrix Multiplication (A × A)

- First row, First column = (85×85) + (78×92) = 14401
- First row, Second column = (85×78) + (78×88) = 13494
- Second row, First column = (92×85) + (88×92) = 15916
- Second row, Second column = (92×78) + (88×88) = 14920

Final result $A^2 = \begin{bmatrix} 14401 & 13494 \\ 15916 & 14920 \end{bmatrix}$

Matrix transpose

original $A = \begin{bmatrix} 85 & 78 \\ 92 & 88 \end{bmatrix}$ Transpose $A^T = \begin{bmatrix} 85 & 92 \\ 78 & 88 \end{bmatrix}$

Determinant $|A| = ad - bc$

$= (85 \times 88) - (78 \times 92)$

$= 7480 - 7176$

$= 304$

Matrix Inverse

$$A^{-1} = \frac{1}{(ad-bc) \times [d \; -b \; ; -c \; a]}$$

$$= \frac{1}{304 \times [88 \; -78 \; ; -92 \; 85]}$$

$$= [88/304 \; -78/304 \; ; \; -92/304 \; 85/304]$$

$$A^{-1} = \begin{bmatrix} 0.28947 & -0.25658 \\ -0.300263 & 0.27961 \end{bmatrix}$$

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Task-7 Line, Plane & Hyperplane

Line - 2D space

$y = mx + c$ where x = math marks
y = physics marks

$St_1 = (85, 78)$ $St_2 = (92, 88)$

Plane - 3D space

- X-axis = math
- Y-axis = Phy
- Z-axis = che

$ax + by + cz + d = 0$

$x + y + z = 240$

Hyperplane

$w_1x_1 + w_2x_2 + \dots + w_5x_5 + b = 0$

- $x_1, x_2 \dots x_5$ = 5 subjects marks
- $w_1, w_2 \dots w_5$ = weights
- b = bais / intercept

Task-8 Covariance

$$cov(x, y) = \sum_{i=1}^{n} \frac{(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

maths variance = 166.52
Physics variance = 138.12
Covariance (maths, Physics) = 144.19

Task-1 Eigenvalues & Eigenvectors

Egienvalues (λ), Eigenvector (v)

Covariance matrix = 5 x 5

- Largest Eigenvalue $\lambda_1 = 685.28$
- 2nd Eigenvalue $\lambda_2 = 19.41$
- 3rd Eigenvalue $\lambda_3 = 10.94$
- 4th Eigenvalue $\lambda_4 = 9.62$
- 5th Eigenvalue $\lambda_5 = 2.16$

Total Variance = 166.52 + 138.12 + 142.35 + 77.67 + 202.74
= 727.41

Variance explain by λ_1 :

$\left(\frac{685.28}{727.41}\right) \times 100 = 94.21\%$